
Technical University of Crete
School of Electrical and Computer Engineering
Course: **Randomized Algorithms**
Exercise 3 (100/1000)
Report Submission Date: December 9, 2024
Instructor: Athanasios P. Liavas

(50) In the first part of this exercise, we shall study the Max-Cut problem in graphs.

1. Simulate the randomized Large-Cut algorithm for random $G_{n,p}$ graphs (for example, $n = 10^2, 10^3, p = 10^{-2}, 10^{-1}, .8$). Repeat the algorithm if you do not get a large-cut, up to $\frac{m}{2} + 1$ times. What do you observe?
2. Simulate the deterministic Large-Cut algorithms, for random $G_{n,p}$ graphs (for example, $n = 10^2, 10^3, p = 10^{-2}, 10^{-1}, .8$).

(50) In the second part of this exercise, we shall study the concentration of the number of empty bins around its mean, in the m -Balls and n -Bins experiment.

1. Throw independently and uniformly at random m balls into n bins (for example, $m = 500, 1000, n = 500, 1000$).
2. Count the number of empty bins.
3. Repeat the experiment K times (for example, $k = 10^3, 10^4$).
4. Compute the arithmetic mean of the number of empty bins and compare it with the statistical mean.
5. Plot the histogram of the number of empty bins in the interval $[n_{\min} : 1 : n_{\max}]$, where $n_{\min}$ is the *smallest possible* number of empty bins and $n_{\max}$ is the *largest possible* number of empty bins (for example, if $m = 100$ and $n = 500$, then $n_{\min} = 0$ and $n_{\max} = 499$). What do you observe concerning the concentration around the mean?